Table 63. Evaluation protocol and release-gate rules

Measure	Definition	Test set and sample	Method and responsibility	Decision rule
Recall@10, MRR@10, nDCG@10	For each answerable question, a hit if any gold passage is in the top 10; Recall@10 is the mean of hits. MRR and nDCG as in Table 28	Private test split: at least 300 answerable questions in v0 (about 150 per language) and 1,200 in v1	Automatic scoring by the independent custodian of the test split; script, outputs, configuration and corpus version archived	At G2 and G3: met if the point estimate meets the threshold and the 95% interval is no wider than plus or minus 5 points (about 4 points at n = 300), otherwise annotate more items; at G4: the lower bound must meet it; safety thresholds judged on the upper bound
Citation precision	Share of sampled answer sentences whose cited passage fully supports the sentence (supported, partly supported or not supported; only supported counts)	400 answer sentences drawn at random from test answers, at least 200 per language	Two trained annotators who do not know the system version; a third adjudicates; Cohen's kappa [121] reported (target 0.70 or more)	Point estimate at least 0.95 (interval about plus or minus 2 points at n = 400)
Unsupported-claim rate	Share of sampled sentences not supported by any retrieved passage	Same sample as citation precision	Same annotators	Point estimate at most 2%
Correct abstention and answer coverage	Abstentions on out-of-scope questions; answers on answerable questions	150 out-of-scope test questions in v0 (200 written: 50 development, 150 test); all answerable test questions	Automatic	Abstention at least 0.90 (interval about plus or minus 5 points); coverage at least 0.80
Language parity	Recall@10 for English minus Recall@10 for Bangla, with a bootstrap interval	Per-language test subsets	Automatic	Reported in v0, where the interval is too wide to decide; release condition from v1 onwards
Robustness	Recall@10 on variant-spelling and romanised queries, on questions answered in OCR-derived documents, and on code-mixed queries	At least 100, 100 and 50 test items respectively	Automatic	Bangla-script variant spelling at least 0.75 in v0; romanised, OCR-derived and code-mixed results reported without a gate in v0; romanised and code-mixed queries gated from v1, once transliteration is delivered (Table 37)
Prompt injection and prohibited outputs	Share of red-team attempts (instructions hidden in documents or queries) that change behaviour; requests for prohibited outputs	At least 100 attempts, refreshed at each release	Security lead and an external red team	At most 5%; any leak of non-public data is a failure
Privacy	Presence of DC2 or DC3 data in logs, outputs or evaluation sets; access-control tests	Log samples and automated scans	Governance analyst; annual independent audit	Zero findings
Latency and availability	p95 response time; share of working hours available	Load test with 30 concurrent users at G2; monitoring during G3	Platform lead	3 s for search and 10 s for answers; availability reported in Year 1 and at least 99% from Year 2 (NFR-04)
Usefulness	Mean rating on a 5-point scale	At least 30 pilot users; at least 100 answers rated by Bangla editors	Product owner and evaluation team	At least 4.0
P2 extraction accuracy	Share of extracted table cells that match the source document	Stratified random sample of at least 1,000 cells across years and ministries	Two independent checkers	At least 99% (interval about plus or minus 0.6 points)
Time to evidence	Median time to find and cite authoritative evidence on matched tasks, with and without the tool	At least 20 analysts, 6 tasks each, crossover design	Evaluation team; independent evaluator at month 18	Effect and interval reported; informs G4, no pass mark at G3
Reproducibility	Re-scoring of archived outputs on the private split	Every release used for a gate decision	Independent evaluator before G4	Re-scored metrics within 1 point of those reported

Intervals are 95% Wilson [122] or bootstrap [123] intervals. BanglaPolicyQA v0 has 500 answerable questions (40% public development, 60% private test) and 200 out-of-scope questions. Sample sizes are minimums; the ERRC and TAB may require more for tier T2 uses.

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